

SDG ASSIGNMENT

Sustainable Development Goal 11 — Sustainable Cities and Communities

Sustainable Development Goal 13 — Climate Action

DESIGN OF A NAÏVE BAYES CLASSIFIER FOR PREDICTING AIR QUALITY LEVELS

A Machine Learning Approach to Environmental Monitoring

Submitted by

K. Vamsi Kumar Reddy

Department of Electronics and Communication Engineering
Saveetha Institute of Medical and Technical Sciences, Chennai

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1. Introduction

Air pollution is among the most pressing environmental and public health challenges of the twenty-first century. Rapid urbanization, vehicular emissions, industrial activity, and construction dust have caused air quality in many cities to deteriorate to levels that pose serious risks to human health and ecosystems. According to global environmental monitoring bodies, poor air quality contributes to millions of premature deaths every year and significantly affects respiratory and cardiovascular health.

Traditional air quality monitoring relies on physical sensor networks that measure pollutant concentrations such as PM2.5, PM10, NO₂, SO₂, CO, and O₃. While accurate, these systems are often expensive to deploy at scale and may not provide timely predictive insight. Machine learning offers a complementary approach: by learning patterns from historical air quality data, a classifier can predict the air quality category (for example, Good, Moderate, Poor, or Hazardous) for new, unseen conditions, enabling early warnings and informed decision-making.

This assignment presents the design of a Naïve Bayes classifier — a simple yet powerful probabilistic machine learning algorithm — for predicting air quality levels based on measurable environmental parameters. The report covers the theoretical foundation, dataset design, step-by-step model construction, a worked numerical example, evaluation strategy, and the classifier's relevance to global sustainability goals.

2. Relevance to Sustainable Development Goals (SDGs)

This project directly supports the United Nations 2030 Agenda for Sustainable Development, particularly the following goals:

SDG	Goal	Connection to this Project
SDG 3	Good Health and Well-being	Early prediction of hazardous air quality helps reduce respiratory and cardiovascular illness by enabling timely public health advisories.
SDG 11	Sustainable Cities and Communities	Supports smart-city air quality monitoring systems that inform urban planning, traffic management, and pollution-control policy.
SDG 13	Climate Action	Continuous classification of air quality trends helps track pollution sources and supports data-driven climate mitigation strategies.
SDG 9	Industry, Innovation and Infrastructure	Demonstrates how low-cost machine learning models can be integrated into IoT-based environmental sensing infrastructure.

3. Problem Statement

Given a set of environmental parameters recorded at a location — such as concentrations of PM2.5, PM10, NO₂, CO, SO₂, and O₃, along with meteorological factors like temperature and humidity — the goal is to design a classification model that can predict the corresponding Air Quality Index (AQI) category. The model must handle multiple input features, work efficiently with limited training data, and provide fast, interpretable predictions suitable for real-time environmental monitoring dashboards.

4. Objectives

1. To study the mathematical principles underlying the Naïve Bayes classification algorithm.
2. To design a Naïve Bayes-based model for classifying air quality into predefined categories.
3. To identify and structure a suitable dataset of air pollutant and meteorological features.
4. To demonstrate the classifier's working through a step-by-step numerical example.
5. To evaluate model performance using standard classification metrics.
6. To relate the project outcomes to the United Nations Sustainable Development Goals.

5. Overview of Naïve Bayes Classifier

Naïve Bayes is a supervised machine learning algorithm based on Bayes' Theorem, used primarily for classification tasks. It is called "naïve" because it makes a simplifying assumption: that every feature used to predict the class is conditionally independent of every other feature, given the class label. Although this assumption rarely holds perfectly in real-world data, the algorithm performs remarkably well in practice, especially for problems involving multiple categorical or continuous features, such as air quality classification.

The classifier calculates the probability of each possible output class (for example, Good, Moderate, Poor, Severe) given the observed input features, and assigns the class with the highest posterior probability. Because it relies on straightforward probability computations rather than iterative optimization, Naïve Bayes is computationally inexpensive, fast to train, and well-suited for real-time or embedded environmental monitoring systems.

5.1 Why Naïve Bayes for Air Quality Prediction?

- Handles multiple continuous features (pollutant concentrations) efficiently using the Gaussian variant.
- Performs well even with relatively small or moderately sized training datasets.
- Computationally lightweight — suitable for deployment on low-power IoT air-quality sensor nodes.
- Provides probabilistic outputs, allowing confidence levels to accompany each prediction.
- Simple to interpret and update as new sensor data becomes available.

6. Mathematical Foundation — Bayes' Theorem

The classifier is built upon Bayes' Theorem, which describes the probability of a class C given observed features X :

$$P(C | X) = [P(X | C) \times P(C)] / P(X)$$

Where:

- $P(C | X)$ — Posterior probability: probability of class C given the feature set X (e.g., probability the air quality is "Poor" given the pollutant readings).
- $P(X | C)$ — Likelihood: probability of observing features X given class C .
- $P(C)$ — Prior probability: overall probability of class C occurring in the dataset.
- $P(X)$ — Evidence: overall probability of observing features X (acts as a normalizing constant).

Since features (PM2.5, PM10, NO₂, CO, temperature, humidity, etc.) are assumed conditionally independent given the class, the joint likelihood simplifies to a product of individual feature likelihoods:

$$P(X | C) = P(x_1 | C) \times P(x_2 | C) \times P(x_3 | C) \times \dots \times P(x_n | C)$$

As pollutant concentrations are continuous numerical values, the Gaussian (Normal distribution) Naïve Bayes variant is used, in which the likelihood of a feature value is computed using the probability density function of the normal distribution:

$$P(x_i | C) = [1 / (2\pi\sigma_c^2)^{0.5}] \times \exp(- (x_i - \mu_c)^2 / (2\sigma_c^2))$$

Here, μ_c and σ_c^2 are the mean and variance of feature x_i computed separately for each class C from the training data. Finally, the predicted class is the one that maximizes the posterior probability:

$$\hat{C} = \operatorname{argmax}_c P(C) \times \prod_i P(x_i | C)$$

7. Dataset Description

A representative dataset for this task consists of daily or hourly environmental readings from air quality monitoring stations. Each record (row) corresponds to one observation, and each column represents a feature or the target label.

Feature	Description	Type / Unit
PM2.5	Fine particulate matter concentration	Continuous — $\mu\text{g}/\text{m}^3$
PM10	Coarse particulate matter concentration	Continuous — $\mu\text{g}/\text{m}^3$
NO ₂	Nitrogen dioxide concentration	Continuous — $\mu\text{g}/\text{m}^3$
SO ₂	Sulfur dioxide concentration	Continuous — $\mu\text{g}/\text{m}^3$

Feature	Description	Type / Unit
CO	Carbon monoxide concentration	Continuous — mg/m ³
O ₃	Ground-level ozone concentration	Continuous — µg/m ³
Temperature	Ambient air temperature	Continuous — °C
Humidity	Relative humidity	Continuous — %
AQI Category (Target)	Air quality class label	Categorical — Good / Moderate / Poor / Severe

Table 1: Feature set used for training the Naïve Bayes air quality classifier.

The target variable, AQI Category, is typically derived from the Central Pollution Control Board (CPCB) or Environmental Protection Agency (EPA) breakpoint standards, which convert raw pollutant concentrations into standardized categories such as Good, Moderate, Poor, Very Poor, and Severe.

8. System Design and Methodology

The end-to-end design of the classifier follows the standard machine learning pipeline, adapted for environmental sensor data:

8.1 Data Collection

Historical pollutant and weather data are gathered from government air quality monitoring stations, open datasets, or IoT-based sensor nodes (e.g., ESP32-based low-cost sensors measuring PM2.5/PM10).

8.2 Data Pre-processing

- Handling missing sensor readings through mean/median imputation.
- Removing outliers caused by sensor malfunction.
- Normalizing continuous features where required.
- Encoding the target AQI category as class labels.

8.3 Splitting the Dataset

The dataset is divided into a training set (typically 70–80%) and a testing set (20–30%), ensuring the class distribution is preserved (stratified split).

8.4 Model Training

For each class C (Good, Moderate, Poor, Severe), the algorithm computes the prior probability $P(C)$ from class frequencies in the training data, and the mean (μ) and variance (σ^2) of each pollutant feature within that class.

8.5 Prediction

For a new observation, the trained model computes the posterior probability for every class using the Gaussian likelihood formula and Bayes' Theorem, then outputs the class with the highest posterior probability as the predicted AQI category.

8.6 System Architecture

Stage 1	Sensor / dataset input → PM2.5, PM10, NO ₂ , SO ₂ , CO, O ₃ , temperature, humidity
Stage 2	Data cleaning and pre-processing module
Stage 3	Feature-wise mean and variance computation per class (training phase)
Stage 4	Gaussian Naïve Bayes probability engine (Bayes' Theorem application)
Stage 5	Output: Predicted AQI category with confidence score

Table 2: Five-stage pipeline of the proposed Naïve Bayes air quality classification system.

9. Working Example — Step-by-Step Calculation

To illustrate the classifier's operation, consider a simplified dataset with a single feature, PM2.5 concentration (µg/m³), and two classes: "Good" and "Poor".

Class	Mean PM2.5 (µ)	Variance (σ²)
Good	35	25
Poor	110	64

Table 3: Class-wise mean and variance of PM2.5 estimated from training data.

Assume the prior probabilities are P(Good) = 0.6 and P(Poor) = 0.4, and a new sensor reading of PM2.5 = 45 µg/m³ is to be classified.

Step 1 — Apply the Gaussian likelihood formula for each class

$$P(x=45 \mid \text{Good}) = [1/(2\pi \times 25)^{0.5}] \times \exp(-(45-35)^2 / (2 \times 25)) \approx 0.0216$$

$$P(x=45 \mid \text{Poor}) = [1/(2\pi \times 64)^{0.5}] \times \exp(-(45-110)^2 / (2 \times 64)) \approx 0.00003$$

Step 2 — Compute posterior scores (unnormalized)

$$\text{Score}(\text{Good}) = P(\text{Good}) \times P(x=45 \mid \text{Good}) = 0.6 \times 0.0216 \approx 0.01296$$

$$\text{Score}(\text{Poor}) = P(\text{Poor}) \times P(x=45 \mid \text{Poor}) = 0.4 \times 0.00003 \approx 0.000012$$

Step 3 — Compare and classify

Since Score(Good) ≈ 0.01296 is much greater than Score(Poor) ≈ 0.000012, the classifier predicts the class "Good" for a PM2.5 reading of 45 µg/m³. In a full implementation, the same procedure is repeated across all six

pollutant features simultaneously, and their individual likelihoods are multiplied together (per the conditional independence assumption) before comparing final posterior scores across all AQI categories.

10. Model Evaluation Metrics

Once trained, the classifier's performance on the held-out test set is assessed using standard classification metrics:

Metric	Description
Accuracy	Proportion of total predictions that were correctly classified.
Precision	Of all instances predicted as a given AQI class, the fraction that were actually correct.
Recall	Of all actual instances of a given AQI class, the fraction correctly identified by the model.
F1-Score	Harmonic mean of precision and recall, useful for imbalanced AQI class distributions.
Confusion Matrix	Tabular breakdown showing correct and incorrect predictions across all AQI categories.

Table 4: Evaluation metrics recommended for validating classifier performance.

11. Advantages and Limitations

11.1 Advantages

- Fast training and prediction, suitable for real-time air quality dashboards.
- Performs well even with a relatively small training dataset.
- Simple to implement and interpret, requiring minimal computational resources.
- Naturally provides probability estimates, useful for confidence-based alerts.
- Scales easily as more pollutant features or monitoring stations are added.

11.2 Limitations

- The conditional independence assumption rarely holds fully — pollutant concentrations (e.g., PM2.5 and PM10) are often correlated.
- Sensitive to skewed or non-Gaussian feature distributions unless the data is properly transformed.
- May underperform compared to ensemble methods (e.g., Random Forest) on highly non-linear pollutant interactions.
- Class imbalance (e.g., "Severe" days being rare) can bias prior probabilities and predictions.

12. Applications in Environmental Sustainability

- Smart city dashboards that provide real-time AQI category alerts to residents.
- Early-warning systems for schools, hospitals, and outdoor event organizers during hazardous pollution episodes.
- Integration with low-cost IoT sensor nodes for hyperlocal air quality mapping in under-monitored regions.
- Supporting policymakers with data-driven evidence for traffic restrictions or industrial emission controls.
- Enabling citizen-facing mobile applications that translate raw sensor data into actionable health guidance.

13. Conclusion

The Naïve Bayes classifier offers a computationally efficient and interpretable approach for predicting air quality levels from pollutant and meteorological data. Despite its simplifying independence assumption, the Gaussian Naïve Bayes model demonstrates strong practical performance for multi-class AQI classification, as shown through the mathematical derivation and worked example in this report. By enabling fast, low-cost, and real-time predictions, such classifiers can be embedded into IoT-based environmental monitoring systems, directly supporting global efforts toward SDG 3 (Good Health and Well-being), SDG 11 (Sustainable Cities and Communities), and SDG 13 (Climate Action). Future work may explore hybrid models that combine Naïve Bayes with ensemble or deep learning techniques to better capture inter-feature correlations while retaining its speed and simplicity.

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